Precision Music Intervention through Computer Vision-Based Human Affect Assessment

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Introduction

Alzheimer's Disease and Related Dementias(ADRD) is a neurological disease which can cause memory loss, cognition and language disorder. It is not curable, but there are some medicines and approaches to slow it down. We propose to use music therapy to decelerate the progression of AD/ADRD. Music has significant potential in improving mood, alleviating anxiety and depression, and also can address symptoms of dementia without using pharmacological methods. However, its widespread applications for mood enhancement and therapeutic effects have not been fully discovered. One major obstacle in expanding access to personalized music intervention is finding a cost-effective solution to tailor music selection/playback according to human emotional response. The objective of this study is to evaluate an AI-based human affect assessment system that can be used to detect and then modify music playback to meet therapeutic objectives for mood modulation with little staffing requirement.

Method

The current stage is to collect specialized dataset containing both facial and body information during people listening to the music.

- Participants: total 32 subjects
  - Young People (18-25) – University of South Florida
  - Older Adults (65+)-- Unisen Senior Living
- Survey types:
  - Personal Information/ Music Preference Survey
  - Emotion Baseline Survey
  - Music-Intervened Emotion Survey
- Target Emotions:
  - Sad, Happy, Angry/Annoyed, Relaxed
  - Others

Procedure

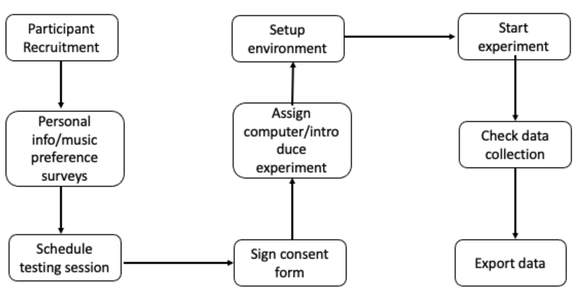


Figure 1: Procedure of Protocol

Experiment Flow Chart

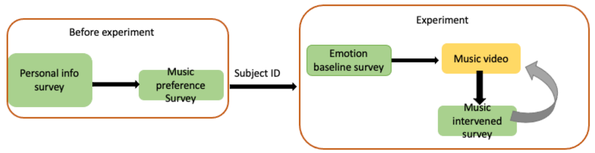


Figure 2: Experiment Flow Chart

Questions	Answers
Have you heard this piece before?	• I have never heard this song before • I think I might have heard this song once to twice before • I am some what familiar with this song • I am very familiar with this song
How would you describe your current emotion?	Happy; Sad; Angry/Annoyed; Relaxed; Other
Please rate the intensity of your current emotion	Mild – moderate – strong
How engaged are you with this music piece?	Mild – moderate – strong

Table 1: Music-Intervened Emotion Survey

Devices Setup

Devices:

- Computer with iMotion software
- Headset
- Web Camera
- Eye Tracking Device

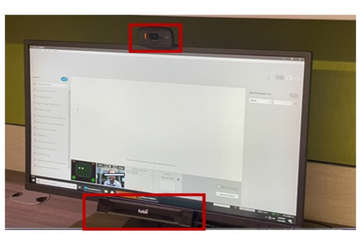


Figure 3: Data Collection Devices Setup

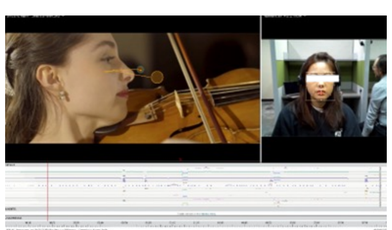


Figure 4: Illustration of Collected Data

Data Distribution

Total 19 subjects from young people group

Emotion	Happy	Sad	Relaxed	Angry/ Annoyed	Other	Total
number	71	19	97	17	24	228

Table 2: Data Distribution for 19 Subjects

Results

Fine Tune High-Speed face Emotion recognition on collected data

Data Size:

- 8 subjects
- 9101 images

Data Splitting:

- Leave one subject out for testing
- Left 7 subjects: 80% training; 20% validation

Current Accuracy: 41.86%

Future Works

- Finish data collection for young people and older adult
- Develop a model that can precisely monitor low intensity human emotions
- Propose new methodologies to develop a music therapy system which can adjust music selection based on the assessed emotion to direct participant's mood positively

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References

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